Printed Pages:02 **Sub Code: KAU -061**

Paper Id: 237016 **Roll No.** | | | | | | | | | | |

B. TECH
(SEM VI) THEORY EXAMINATION 2022-23
AUTOMOTIVE ELECTRICAL AND ELECTRONICS

Time: 3 Hours ***Total Marks: 100***

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. 2 x 10 = 20

(a) What are the benefits of wiring harness?

(b) State Kirchhoff's law.

(c) Briefly explain the working of rectifier.

(d) State two advantages and two disadvantages of a permanent magnet starter.

(e) Write short note on ballast resistance.

(f) Draw a circuit diagram of a oil pressure warning light.

(g) Write any four benefits of electrically driven air conditioning system.

(h) What do you understand by micromechanics?

(i) Briefly discuss about FlexRay.

(j) What is an optoelectronic sensor?

SECTION B

2. Attempt any *three* of the following: 10x3=30

(a) Compare positive earthing to negative earthing system? Which one is preferred and why? Explain in detail.

(b) Describe the working of balancing coil type of fuel gauge with neat sketch.

(c) Discuss in detail about the various methods for adjustment of headlights.

(d) Explain about bus system? Also mention its advantages & requirements.

(e) Discuss about the following:
  (i) Pressure Sensors
  (ii) Position Sensors

SECTION C

3. Attempt any *one* part of the following: 10x1=10

(a) What do you understand by battery ratings? Explain any two types of commonly used battery ratings.

(b) Write a note on following:
  (i) Principle of operation of lead -acid battery
  (ii) Sulphation failure

QP23EP1_311 | 28-06-2023 09:12:41 | 132.154.2.188

QP23EP1_311 | 28-06-2023 09:12:41 | 132.154.2.188

4. Attempt any *one* part of the following: 10x1=10

(a) What is an Alternator? Compare it with Generator. Which one is preferred? Explain the reasons in detail.

(b) Write the name of various types of starters? Explain any two in detail.

5. Attempt any *one* part of the following: 10x1=10

(a) Explain the working of ignition system with ignition coil of 4-cylinderSI engine.

(b) Discuss the important characteristics of ideal spark plug.

6. Attempt any *one* part of the following: 10x1=10

(a) Explain ABS control unit in detail.

(b) Describe the following: (i) ECM (ii) Steering control unit

7. Attempt any *one* part of the following: 10x1=10

(a) Discuss in detail about acceleration & vibration sensors.

(b) Write the note on the following:
   (i) Stepper Motor
   (ii) Immobilizer System

QP23EP1_311 | 28-06-2023 09:12:41 | 132.154.2.188

QP23EP1_311 | 28-06-2023 09:12:41 | 132.154.2.188